Lab 4: Group Policy Objects (GPO) – Policy Enforcement

Objective

Demonstrate the creation, configuration, deployment, and verification of a Group Policy Object (GPO) in an Active Directory environment. This lab simulates a real-world scenario where administrators restrict user access to system settings.

Environment

- Hypervisor: VirtualBox
- Domain Controller: Windows Server (AD DS)
- Client Machine: Windows 10 (LAB-CLIENT01)
- Domain: LAB (lab.local)
- Domain User: LAB\joejane
- Administrator Account: LAB\Administrator

Scenario

In a corporate environment, IT administrators often restrict access to system settings such as Control Panel to prevent unauthorised configuration changes by standard users.

This lab enforces a policy that blocks Control Panel and PC Settings access for domain users.

Step 1: Open Group Policy Management

1. Logged into the Domain Controller as **LAB\Administrator**
2. Opened **Server Manager → Tools → Group Policy Management**

Step 2: Create a New GPO

1. Navigated to:

```
Forest: lab.local
 └── Domains
      └── lab.local
```

2. Right-clicked the domain and selected **Create a GPO in this domain, and Link it here**

3. Named the GPO:

```
LAB-Disable-ControlPanel
```

The GPO was successfully created and linked to the domain.

Step 3: Edit the GPO

1. Right-clicked **LAB-Disable-ControlPanel**
2. Selected **Edit** to open Group Policy Management Editor
3. Navigated to:

```
User Configuration
 └── Policies
     └── Administrative Templates
         └── Control Panel
```

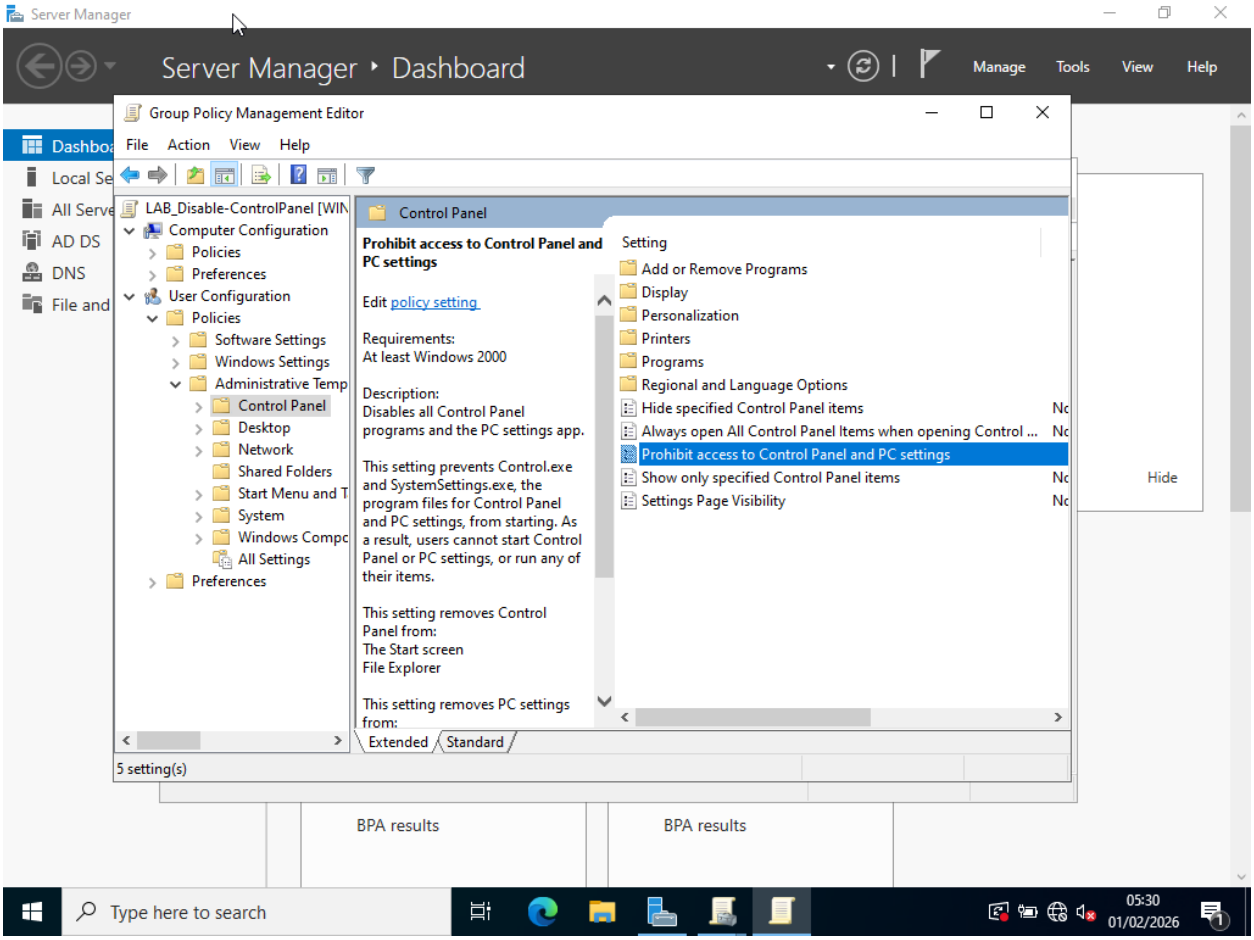
Step 4: Configure Policy Setting

1. Opened the policy:

```
Prohibit access to Control Panel and PC settings
```

2. Set the policy state to **Enabled**
3. Applied and saved the configuration

Screenshot: Lab4_GPO_Disable_ControlPanel.png



Step 5: Apply Group Policy on Client

1. Logged into **LAB-CLIENT01** as **LAB\joejane**
2. Opened Command Prompt
3. Ran:

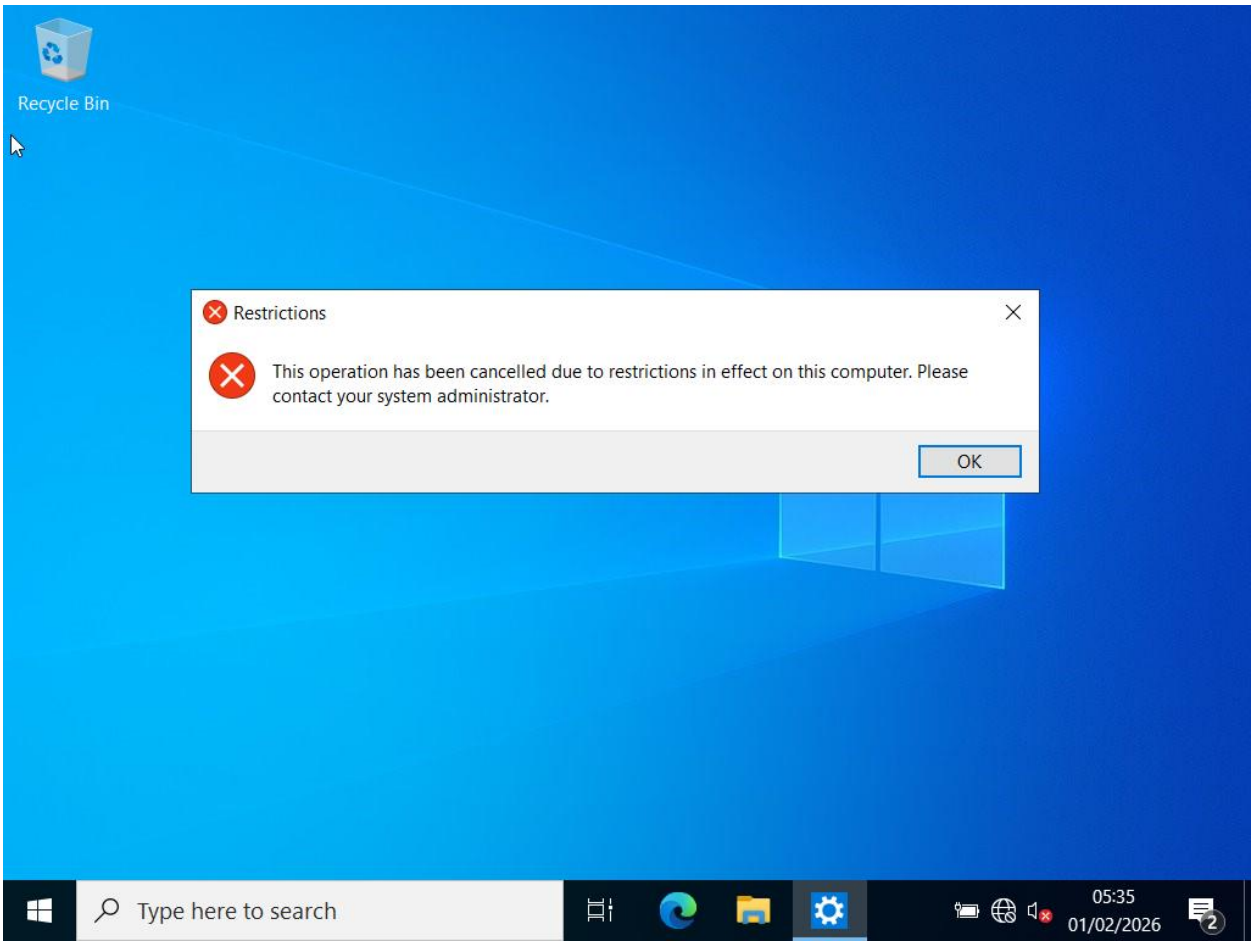
```
gpupdate /force
```

4. Confirmed successful user policy update

Step 6: Verify Policy Enforcement

1. Attempted to open Control Panel on the client machine
2. System displayed a restriction message indicating access was blocked by policy

Screenshot: Lab4_ControlPanel_Blocked.png



Result

The Group Policy Object was successfully applied. The domain user was prevented from accessing Control Panel and PC Settings, confirming correct GPO deployment and enforcement.

Troubleshooting Notes

- Ensured GPO was linked to the correct domain
- Verified user was logged in with a domain account
- Forced policy update to ensure immediate application

Lessons Learned

- GPOs allow centralised control of user and computer behaviour
- User Configuration policies affect domain users regardless of machine

- Group Policy enforcement can be verified immediately using gpupdate
- GPO troubleshooting requires correct scope and linkage

Skills Demonstrated

- Group Policy Object creation and management
- Policy scoping and enforcement
- Active Directory administration
- Client-side policy verification
- Enterprise security controls

Next Steps

- Lab 5: Event Viewer & Security Log Analysis
- Lab 6: Advanced GPOs (Password Policies, Lock Screen Restrictions)

Note on screenshots: Insert screenshots manually using the filenames referenced above.